

Question 1 [15 marks]

Convert the following numbers to the bases indicated.

Original Number	Converted Number
126 (Decimal)	Binary= 111 1110
21 (Hexadecimal)	Octal= 041
4 (Decimal)	Gray code (in 4 bits) = 0110
01101111 (Binary)	Hexadecimal = 6F
20 (Decimal)	BCD = 0010 0000

Question 2 [10 marks]

Perform the following addition using 2's complement binary. Express your answer in both Binary and Decimal. [10 marks]

$$(-10_{10}) + (-14_{10})$$

$$\begin{array}{r} \\ 1 \\ 110110 \quad (-10) \\ 110010 \\ \hline \boxed{1}101000 \quad (-24) \\ \downarrow \downarrow \downarrow \text{2's compl.} \\ 011000 \quad (24) \end{array}$$

$$\begin{array}{l} 10 = 001010 \\ -10 = 110110 \\ 14 = 001110 \\ -14 = 110010 \end{array}$$

Binary: 101000

Decimal: -24

Question 3 [15 marks]

What is the decimal value of the floating point number represented by the following IEEE 754 floating point format?

0	1000 0000	110 0000 0000 0000 0000 0000
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$$\begin{array}{l} \downarrow \qquad \qquad \downarrow \\ 128 - 127 = 1 \qquad 1.11 \end{array}$$

[15 marks]

$$1.11 \times 2^1 = 11.1_2 = 3.5$$

Ans = 3.5

Question 4 [15 marks]

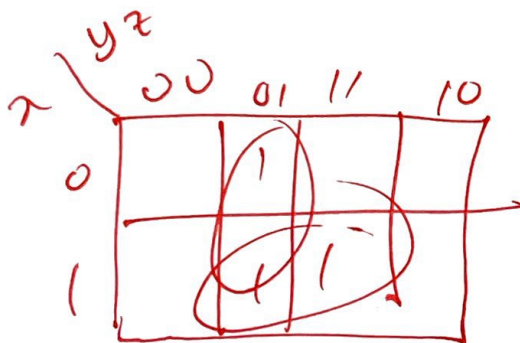
(a) Complete the truth table for the function, $F = xy'z + x'y'z + xyz$

[5 marks]

X	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

(b) Simplify the function in part (a) using Karnaugh map.

[10 marks]



$$F = y'z + xz$$

Ans: $F =$

$$y'z + xz$$

Question 5 [15 marks]

- (a) Simplify the function $F = (A' + B)' + B(A' + AC) + ABC'$ using Boolean Algebra. [10 marks]

$$\begin{aligned} F &= (A' + B)' + B(A' + AC) + ABC' \\ &= AB' + A'B + ABC + ABC' \quad \text{De Morgan's} \\ &= AB' + A'B + AB(C + C') \\ &= AB' + A'B + AB \quad \xrightarrow{\text{alternative}} AB' + (A' + A)B \\ &= A(B + B') + A'B \\ &= A + A'B \\ &= (A + A')(A + B) \quad \text{Distrib. Law.} \\ &= A + B \end{aligned}$$

$F =$ $A + B$

- (b) Use De Morgan's theorem to find the complement of $F = x(y'z' + yz)$ [5 marks]

$$\begin{aligned} F' &= [x(y'z' + yz)]' \\ &= x' + (y'z' + yz)' \\ &= x' + (y + z)(y' + z') \end{aligned}$$

$F' =$ $x' + (y + z)(y' + z')$

Question 6 [20 marks]

Design a circuit that determines the odd parity of a 3-bit input A, B, and C. The circuit outputs a single-bit value P, where $P = 1$ if the number of 1's in the input is even, and $P = 0$ otherwise.

(a) Construct the truth table for the parity checker.

[10 marks]

A	B	C	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

(b) Derive the minimal sum of products expression for the output P. [10 marks]

		BC			
		00	01	11	10
A	0	1		1	
	1		1	1	1

$$P = A'B'C' + A'BC + AB'C + ABC'$$

Question 7 [10 marks]

A sequential circuit has two JK flip-flops, A and B, and one input X. Part of the state diagram is given in the state table below. Determine the flip-flop inputs J_A , K_A , J_B , and K_B .
[10 marks]

Initial State		Input	Next State		Flip Flop Inputs			
A	B	X	A	B	J_A	K_A	J_B	K_B
0	0	0	0	1	0	X	1	X
0	0	1	1	0	1	X	0	X
0	1	0	0	1	0	X	X	0
0	1	1	0	0	0	X	X	1

Excitation Table (JK)

$Q(t)$	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0